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Variable inlet guide vane apparatus and compressor including same

Abstract

A variable inlet guide vane apparatus for imparting a motion to a fluid flow entering into a compressor. The variable inlet guide vane apparatus includes a housing defining a fluid flow passageway. The housing includes a first and second housing portion, a ring gear, and a plurality of guide vanes. The guide vanes include a stem, a gear, and a vane. The stem is disposed between the first and second housing portions, the vane is arranged within the fluid flow passageway, and the gears are arranged exterior the housing.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application in a continuation of U.S. patent application Ser. No. 17/656,260 filed on Mar. 24, 2022, entitled “Variable Inlet Guide Vane Apparatus and Compressor Including Same” hereby incorporated by reference.

FIELD OF THE DISCLOSURE

(1) The field of the disclosure relates generally to compressors, and more particularly, to variable inlet guide vane apparatuses for use with compressors.

BACKGROUND

(2) Inlet guide vane devices can be used to regulate pressure and direction of fluid flow at the inlet of a compressor, such as a centrifugal compressor. Conventional inlet guide vane devices include a plurality of guide vanes arranged circumferentially about a fluid flow path. The plurality of vanes impart a swirling motion to the fluid flow directing the fluid flow at suitable angles entering the compressor to improve efficiency and performance. The plurality of guide vanes may be rotated to adjust the orientation of the guide vanes relative to the fluid flow path to meet air intake requirements of the compressor for various operating conditions.

(3) Each of the plurality of guide vanes may be rotatable relative to a housing mounted in proximity to the inlet of the compressor. High wear components of known inlet guide vane devices may be difficult to access for inspection and/or maintenance. Known inlet guide vane devices have one or more bearings which require maintenance, such as relubrication and/or periodic replacement, requiring downtime of the compressor. In some instances, the guide vane housing must be disconnected from the compressor, and then the guide vane housing disassembled during repair or inspection to access the guide vanes. The disassembly process can result in significant downtime of the compressor. Additionally, disassembly of the housing to access the guide vanes increases the likelihood of handling and installation damage, reducing the overall life of the inlet guide vane device. For example, disassembly of the housing may expose bearing surfaces to contaminants and debris potentially leading to etching and corrosion. Reassembling of the plurality of guide vanes relative to the housing may also result in misalignment.

(4) This section is intended to introduce the reader to various aspects of art that may be related to various aspects of the disclosure, which are described and/or claimed below. This discussion is believed to be helpful in providing the reader with background information to facilitate a better

understanding of the various aspects of the present disclosure. Accordingly, it should be understood that these statements are to be read in this light, and not as admissions of prior art.

SUMMARY

(5) In one aspect an inlet guide vane apparatus includes a housing assembly defining a fluid flow passageway. The housing assembly includes a first housing portion and a second housing portion connected to and positioned axially upstream of the first housing portion. The first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway. A ring gear is rotatable relative to the housing assembly. A plurality of guide vanes is connected to the housing assembly. Each of the guide vanes includes a stem extending from a first end to a second end. At least a portion of the stem is disposed within one of the guide vane openings. A vane gear is disposed at the first end of the stem and operably connected to the ring gear. The vane gear is disposed at an exterior of the housing assembly and a vane is disposed at the second end of the stem and within the fluid flow passageway of the housing assembly. Each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.

(6) In another aspect, a compressor includes a compressor housing including an inlet and a driveshaft rotatably supported within the compressor housing. The compressor further includes an impeller connected to the driveshaft and operable to impart kinetic energy to incoming refrigerant gas upon rotation of the driveshaft. An inlet guide vane apparatus is connected to the compressor inlet. The inlet guide vane apparatus includes a housing assembly defining a fluid flow passageway. The housing assembly includes a first housing portion and a second housing portion connected to and positioned axially upstream of the first housing portion. The first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway. A ring gear is rotatable relative to the housing assembly and a plurality of guide vanes is connected to housing assembly. Each of the guide vanes includes a vane gear operably connected to the ring gear and disposed at an exterior of the housing assembly and a vane disposed within the fluid flow passageway of the housing assembly. Each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.

(7) In yet another aspect, a method of assembling an inlet guide vane apparatus including a housing assembly and a plurality of guide vanes is provided. The housing assembly includes a first housing portion and a second housing portion. Each of the guide vanes includes a stem extending from a first end to a second end. A vane gear is disposed at the first end of the stem and a vane disposed at the second end of the stem. The method includes rotatably connecting a ring gear to the first housing portion and connecting the first housing portion to the second housing portion such that the second housing portion is positioned axially upstream of the first housing portion, such that the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly. The method further includes positioning the plurality of guide vanes relative to the second housing portion such that, when the first housing portion is connected to the second housing portion, at least a portion of the stem of each guide vane is disposed within one of the guide vane openings and the vane gear of each guide vane is positioned at an exterior of the housing assembly. The method includes operably connecting the vane gear of each guide vane to the ring gear.

(8) Various refinements exist of the features noted in relation to the above-mentioned aspects of the present disclosure. Further features may also be incorporated in the above-mentioned aspects of the present disclosure as well. These refinements and additional features may exist individually or in any combination. For instance, various features discussed below in relation to any of the illustrated embodiments of the present disclosure may be incorporated into any of the above-described aspects of the present disclosure, alone or in any combination.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of an example a variable inlet guide vane apparatus including a first and second housing portion, a bearing, a ring gear, and a plurality of guide vanes.
- (2) FIG. 2 is a side view of the variable inlet guide vane apparatus shown in FIG. 1.
- (3) FIG. 3 is an exploded view of the variable inlet guide vane apparatus shown in FIG. 1.
- (4) FIG. 4 is a perspective view of the second housing portion for use with the variable inlet guide vane apparatus shown in FIG. 1.
- (5) FIG. 5 is a rear or downstream view of the second housing portion shown in FIG. 4.
- (6) FIG. 6 is a sectional view of the second housing portion shown in FIG. 4.
- (7) FIG. 7 is a perspective view of the plurality of guide vanes for use with the variable inlet guide vane apparatus shown in FIG. 1.
- (8) FIG. 8 is a side view of one of the plurality of guide vanes shown in FIG. 7.
- (9) FIG. 9 is a top view of the guide vane shown in FIG. 8.
- (10) FIG. 10 is an exploded view of the guide vane shown in FIG. 8.
- (11) FIG. 11 is a perspective view of the first housing portion for use with the variable inlet guide vane apparatus shown in FIG. 1.
- (12) FIG. 12 is a front or upstream view of the first housing portion shown in FIG. 11.
- (13) FIG. 13 is a side view of the first housing portion shown in FIG. 11.
- (14) FIG. 14 is a perspective view of the ring gear for use with the variable inlet guide vane apparatus shown in FIG. 1.
- (15) FIG. 15 is an end view of the ring gear shown in FIG. 14.
- (16) FIG. 16 is a sectional view of the ring gear shown in FIG. 14.
- (17) FIG. 17 is a perspective view of the bearing of the variable inlet guide vane apparatus shown in FIG. 1.
- (18) FIG. 18 is a rear or downstream view of the variable inlet guide vane apparatus having the first housing portion, ring gear, and bearing disconnected, illustrating the plurality of guide vanes arranged within a second channel on the second housing portion.
- (19) FIG. 19 is a perspective view of an assembled compressor for use with the variable inlet guide vane apparatus.
- (20) FIG. 20 is a cross-sectional view of the compressor of FIG. 19 taken along line 2-2.
- (21) Corresponding reference characters indicate corresponding parts throughout the drawings.

DETAILED DESCRIPTION

(22) FIG. 1 is a perspective view of an example variable inlet guide vane apparatus, generally indicated at **100**. The variable inlet guide vane apparatus **100** (also referred to herein as the inlet guide **100**) is suitable for use with compressors, such as centrifugal compressors (see, e.g., compressor **300**, shown in FIGS. **19** and **20**), and can improve the operating range and efficiency of the compressor by imparting a pre-swirl motion to a fluid flow **F** entering the compressor. The inlet guide **100** may be mounted in proximity to an inlet of the compressor and the fluid flow **F** exits the inlet guide **100** with a pre-swirl and enters into the inlet of the compressor, such that the fluid flow **F** contacts an impeller of the compressor with a suitable direction. Alternatively, and/or additionally, the inlet guide vane **100** is mounted in proximity to the inlet for each stage of a multi-stage compressor.

(23) FIG. 2 is a side view of the inlet guide **100**, and FIG. 3 is an exploded view of the inlet guide **100**. In the illustrated embodiment, the inlet guide **100** includes a first housing portion **102** and a second housing portion **104**. The first housing portion **102** and the second housing portion **104** can be connected to form a vane housing assembly **106**. The second housing portion **104** is positioned axially upstream from the first housing portion **102** relative to the direction of fluid flow. The inlet

guide **100** also includes a ring gear **108** rotatably connected to the housing assembly **106**. In the illustrated embodiment, the ring gear **108** is rotatably connected to the first housing portion **102**. The ring gear **108** can be rotatably connected to the housing assembly **106** by a bearing, such as bearing **110** (FIG. 3). In other embodiments, the bearing **110** may be omitted.

(24) Inlet guide **100** further includes a plurality of a guide vanes **112**. Each guide vane **112** is rotatable relative to the vane housing assembly **106** and is operably connected with the ring gear **108** such that rotation of the ring gear **108** causes each of the plurality of guide vanes **112** to rotate in unison. Each of the guide vanes **112** is rotatable relative to the housing assembly **106** such that the orientation of the vane **160** within a fluid flow passageway P of the housing assembly **106** is selectively adjustable. In some embodiments, the guide vanes **112** are rotatable relative to the housing assembly **106** in unison

(25) The inlet guide **100** can also include one or more motors **174** operably connected to one or more of the guide vanes **112** to selectively rotate the guide vanes **112**. The illustrated inlet guide **100** includes a motor **174** mounted to a motor mount on the second housing portion **104**. The motor **174** is operably coupled to one of the guide vanes **112**, also referred to as a drive guide vane, by a driveshaft **175**. As described further herein, rotation of the drive guide vane by the motor **174** causes rotation of the ring gear **108**, which in turn causes rotation of the other guide vanes **112**, also referred to as follower guide vanes. The motor **174** can include any suitable motor that enables the inlet guide **100** to function as described herein, including, for example and without limitation, electric motors. In some embodiments, the motor **174** is a stepper motor. In other embodiments, the motor **174** includes a synchronous motor that operates based on rotation feedback from one or more of the follower vanes, e.g., using a potentiometer or position sensor coupled to one or more of the follower vanes, to ensure all vanes are driven to the desired angle. The motor **174** may include for example and without limitation a DC motor, brushed or brushless, and a synchronous AC motor.

(26) With additional reference to FIG. 3, the first housing portion **102** includes a first annular wall **126** having a first inner surface **128** and a first outer surface **130**. The second housing portion **104** includes a second annular wall **120** having a second inner surface **122** and a second outer surface **124**. The first inner surface **128** and the second inner surface **122** defines the boundary of a fluid flow passageway P extending through the vane housing assembly **106**. The vane housing assembly **106** has a housing axis A.sub.106 extending through the fluid flow passageway P. The first housing portion **102** defines an exit or outlet **132** of fluid flow passageway P, and the second housing portion **104** defines an inlet **134** of the fluid flow passageway P. The fluid flow F enters the vane housing assembly **106** at the inlet **134**, passes through the fluid flow passageway P, and leaves the vane housing assembly **106** at the outlet **132**. The fluid flow F flows through the fluid flow passageway P in a direction that is generally parallel to the housing axis A.sub.106. Fluid flow F leaving the exit **132** has a pre-swirl imparted by the plurality of the guide vanes **112**, as described in further detail herein.

(27) The vane housing assembly **106** includes an exterior area **136** surrounding the first outer surface **130** and the second outer surface **124** and located generally radially outward from the housing assembly **106**. In the illustrated embodiment, at least a portion of each of the plurality of guide vanes **112** is disposed between the first housing portion **102** and second housing portion **104**, and at least a portion of the guide vane **112** and the ring gear **108** is arranged in the exterior area **136** of the vane housing assembly **106**. Accordingly, the ring gear **108** and at least a portion of the guide vane **112** are accessible (e.g., to an operator or technician) for inspection and/or repairs without requiring inlet guide **100** to be disassembled. By way of example, an operator or technician can access the ring gear **108** and a portion of the guide vanes **112** (e.g., vane gears, described in more herein) without first disconnecting the first housing portion **102** from the second housing portion **104**.

(28) FIG. 4 is a perspective view of the second housing portion **104**. FIGS. 5 and 6 are a rear view

and a sectional view, respectively, of the second housing portion **104**. The second annular wall **120** includes a downstream surface **140** and an upstream surface **142**. The downstream surface **140** is generally annular in shape. The second annular wall **120** may have a width $W_{sub.120}$ extending between the second outer surface **124** and the second inner surface **122**. (FIG. 5). The second annular wall **120** may have a height $H_{sub.120}$ extending between the downstream surface **140** and the upstream surface **142**. (FIG. 6). The second inner surface **122** defines the boundary of a second fluid flow passageway $P_{sub.120}$ that includes a diameter $D_{sub.120}$ defined by the second inner surface **122**. In the illustrated embodiment, the second fluid flow passageway $P_{sub.120}$ is generally conical in shape and the diameter $D_{sub.120}$ decreases in a direction from the inlet **134** to the outlet **132** along the axis $A_{sub.106}$. In other embodiments, the second fluid flow passageway $P_{sub.120}$ can be generally cylindrical in shape and the diameter $D_{sub.120}$ defined by the second inner surface **122** is generally constant. The second fluid flow passageway $P_{sub.120}$ has a length $L_{sub.120}$. In the illustrated embodiment, the length $L_{sub.120}$ corresponds to the height $H_{sub.120}$ of the second annular wall **120**. The housing axis $A_{sub.106}$ extends through the second fluid flow passageway $P_{sub.120}$. The upstream surface **142** is generally planar and may be mounted to a compressor. Alternatively, the second housing portion **104** may be mounted to any suitable structure in proximity to the inlet of a compressor. The dimensions of the compressor, e.g., width $W_{sub.120}$, height $H_{sub.120}$, diameter $D_{sub.120}$, and length $L_{sub.120}$, may be scaled to the size of the compressor and the aerodynamic needs of the compressor.

(29) With reference to FIGS. 4-6, the second housing portion **104** includes one or more flanges **144** extending radially outward from the second annular wall **120**. The flange **144** may extend generally perpendicular to the second annular wall **120**. The flanges **144** include one or more fastener openings **146** for receiving suitable fasteners (e.g., screws, bolts, etc.) to connect the second housing portion **104** to a compressor.

(30) With reference again to FIG. 5, the downstream surface **140** includes a plurality of second channel surfaces **148**, each defining a corresponding second channel **150**. Each of the plurality of second channel surfaces **148** are arranged in a radially symmetric pattern about the housing axis $A_{sub.106}$. In the illustrated embodiment, the downstream surface **140** of the second annular wall **120** includes ten second channel surfaces **148** defining ten second channels **150** arranged in a radially symmetric pattern about the housing axis **A 106**. In alternative embodiments, the downstream surface **140** may include any number of second channels surfaces **148** that enables the inlet guide **100** to function as described herein. For example, in some embodiments, there are six second channel surfaces **148** defining six second channels **150**.

(31) In the illustrated embodiment, each of the plurality of second channel surfaces **148** are identical, having the same size and shape. In the illustrated embodiment, the second channel surfaces **148** are in the shape of a segment of a cylindrical surface. Accordingly, the second channels **150** are generally in the shape of a half-cylinder. The second channel surface **148** has a second channel length $L_{sub.148}$ extending from the second inner surface **122** to the second outer surface **124**. In the illustrated embodiment, the second channel surfaces **148** and second channels **150** extend through the entire width $W_{sub.120}$ of the second annular wall **120**. In other embodiments, the second channel surfaces **148** extend only partially through the width $W_{sub.120}$ of the second annular wall **120**. The second channel surfaces **148** are sized and shaped such that second channels **150** are sized and shaped to receive at least a portion of the guide vanes **112** therein, as described in further detail herein.

(32) In the illustrated embodiment, each second channel surface **148** includes a secondary channel surface **152** defining a slot **154**. The secondary channel surface **152** extends radially from the second channel surface **148**, such that the slot **154** has a depth $D_{sub.152}$ extending from the second channel surface **148** (FIG. 4). The secondary channel surface **152** defining the slot **154** includes a first end **156** and a second end **158**, and a secondary channel length $L_{sub.152}$ extending therebetween (FIG. 5). The secondary channel surface further defines a slot width $W_{sub.152}$.

(33) FIG. 7 is a perspective view of the plurality of guide vanes **112**. Each of the plurality of guide vanes **112** includes a vane **160**, a stem **162**, and a vane gear **164**. The plurality of guide vanes **112** are arranged in a radially symmetric pattern mirroring the radially symmetric pattern of the plurality of second channel surfaces **148** on the second housing portion **104**. In the illustrated embodiment, there are ten guide vanes **112** corresponding to the ten second channels surfaces **148**. In other embodiments, the inlet guide **100** can include any suitable number of guide vanes **112** that enables the inlet guide **100** to function as described herein. For example, in some embodiments there may be six guide vanes **112** corresponding to six second channel surfaces **148**.

(34) In the illustrated embodiment, the vane **160** is formed integrally with the stem **162**. For example, the vane **160** and the stem **162** may be molded as a single piece. In alternative embodiments, the vane **160** may be formed separately from and connected to the stem **162**. The vane **160** is substantially triangular, in the illustrated embodiment, and includes a first vane side **166** and an opposing second vane side **168**. The first and second vane sides **166**, **168** are substantially planar. The first and second vane sides **166** and **168** connect at a trailing edge **170** and a leading edge **172**. The trailing and leading edges **170** and **172** may be knife-like in shape. The plurality of vanes **160** are selectively rotatable to selectively occlude or block, fluid flow **F** through the fluid flow passageway **P.sub.120**. The plurality of vanes **160** may be prevented from being rotated to arrange the vanes **160** in a closed position, in which the trailing edge **170** of each guide vane is in contact with, or in close proximity with, the leading edge **172** of an adjacent guide vane, as described herein.

(35) The first and second housing portions **102** and **104** cooperatively define a plurality of guide vane passages **P.sub.160**. Each of the plurality of vanes **160** may cover one of the vane passages **P.sub.160**. Each vane passage **P.sub.160** is a portion of the fluid flow passageway **P**. The vane passages **P.sub.160** are spaced circumferentially about the fluid flow passageway **P**. The vane **160** is any shape or size enabling the inlet guide **100** to function as describe herein. Additionally, the shape and size of the vane **160** may be selected based upon the intended application of the inlet guide **100**. For example, the size, shape, and angle of the vane **160** may be selected based on the type and configuration of the compressor, the operating conditions, and/or the fluid type used with the compressor. Each of the guide vanes **112** is rotatable relative to the vane housing assembly **106** such that the orientation of the vane **160** within the fluid flow passageway **P** is selectively adjustable.

(36) FIGS. **8** and **9** are side and top views, respectively, of one of the guide vanes **112** shown in FIG. **7**. The stem **162** extends along a stem axis **A.sub.162** from a first, inner end **178** to a second, outer end **176**. The vane **160** is disposed at the first, inner end **178** of the stem **162**, and the vane gear **164** is disposed at the second, outer end **176** of the stem **162**. The stem **162** has a stem length **L.sub.162** between the inner end **178** and the outer end **176**. The stem length **L.sub.162** may be substantially similar to the second channel length **L.sub.148**. When the stem **162** is arranged within one of the plurality of second channels **150**, the stem axis **A.sub.162** is perpendicular to the housing axis **A.sub.160**.

(37) In this illustrated embodiment, the guide vane **112** includes a stop **177** extending circumferentially about the stem **162**, and radially outward in a direction generally perpendicular to the stem axis **A.sub.106**, from the stem **162**. The stop **177** is sized and shaped to fit within the slot **154** defined by the secondary channel surface **152**. In the illustrated embodiment, the stop **177** is generally rectangular in shape. The stop **177** and the secondary channel surface **152** are sized and shaped such that the stop **177** slides along the length **L.sub.152** relative to the secondary channel surface **152** when the guide vane **112** rotates relative to the first housing portion **102** and the second housing portion **104** about the stem axis **A.sub.162**. In addition, the slot width **W.sub.152** is sized and shaped such that the stop **177** contacts the secondary channel surface **152**, preventing the guide vane **112** from translating along a direction parallel to the stem axis **A.sub.162**. The slot width **W.sub.152** is sized to provide sufficient clearance between the stop **177** and the secondary channel

surface **152** such that the stop **177** may translate along the slot length L.sub.152.

(38) The stop **177** extends only partially around the circumference of the stem **162**. In the illustrated embodiment, the stop **177** extends circumferentially an arc angle of about 45° around the circumference of the stem **162**. In other embodiments, the stop **177** may extend an arc angle greater than or less than 45°. For example, in some embodiments, the stop **177** may extend an arc angle of 90° around the stem **162**. In another example, the stop **177** may extend an arc angle of 30° around the stem **162**.

(39) The stop **177** limits rotation of the guide vane **112** about the stem axis A.sub.162 when the stop **177** engages with a stop surface of at least one of the first housing portion **102** or the second housing portion **104**, as described further herein.

(40) The vane gear **164** includes a top surface **180**, a bottom surface **182**, and a wall **184** extending between the top surface **180** and the bottom surface **182**. The wall **184** is generally cylindrical in shape (FIG. 8). The vane gear **164** includes a plurality of gear teeth **186** extending radially outward from the wall **184**. In some embodiments, such as the illustrated embodiment, the vane gear **164** is a tapered gear—i.e., the gear teeth **186** are tapered or angled radially inward or outward from the top surface **180** to the bottom surface **182**. In some embodiments, the vane gear **164** may be a bevel gear. In some embodiments, the vane gear **164** may be a helical gear.

(41) In the illustrated embodiment, the vane gear **164** is a partial gear in which the gear teeth **186** extend around only a portion of the wall **184**. In the illustrated embodiment, the gear teeth **186** extend an arc angle of approximately 225° around the wall **184**. In other embodiments, the gear teeth **186** may extend around the wall **184** an arc angle of greater than or less than 225°. Further, in the illustrated embodiment, the gear teeth **186** are positioned generally opposite the stop **177** on the stem **162**. Accordingly, during operation, the vane gear **164** is arranged to engage with the ring gear **108**, while the stop **177** is captured within the slot **154**.

(42) FIG. 10 is an exploded view of the guide vane **112**. In some embodiments, the vane gear **164** is removably connected to the stem **162** of the guide vane **112**. In the illustrated embodiment, for example, the outer end **176** of the stem **162** is received within a central opening of the vane gear **164** to connect the vane gear **164** to the stem **162**. In some embodiments, the stem **162** and the vane gear **164** may be connected using a press-fit engagement when the outer end **176** is disposed within the central opening of the vane gear **164**. Additionally, and/or alternatively, an epoxy, or other suitable adhesive, may be used to connect the stem **162** and the vane gear **164**. Additionally, in the illustrated embodiment, the stem **162** includes a key **188** disposed at the outer end **176** and the vane gear **164** includes a keyed boundary **190** defining a keyed opening **189** sized and shaped to receive the key **188**. When the key **188** is disposed within the keyed opening **189**, the stem **162** and the vane gear **164** are frictionally engaged and rotation of the vane gear **164** is transmitted to the stem **162**. In some embodiments, the key **188** may be press fit within the keyed opening **189**. In some embodiments, the key **188** may include a channel, and the boundary **190** may include a key that is sized and shaped to fit within the channel. Alternatively, and/or additionally, the key **188** and the boundary **190** may include any suitable features that enable frictional engagement of the vane gear **164** and the stem **162**. In other embodiments, the vane gear **164** is integral with the guide vane (e.g., the vane gear **164** is formed integrally with the stem **162**).

(43) In the illustrated embodiment, the guide vane **112** includes an alignment feature **191**. The alignment feature **191** receives a portion of the driveshaft of a motor, such as driveshaft **175** of motor **174**, enabling the guide vane **112** to be operably connected to the motor. Accordingly, the alignment feature **191** is shaped complimentary to the driveshaft. For example, the alignment feature **191** may be keyed, semicircular, or star shaped, or any suitable shape such that the alignment feature **191** mates with the driveshaft in order to frictionally couple the driveshaft to the alignment feature **191**. Alternatively, and/or additionally, the alignment feature **191** may be sized and shaped to receive an alignment tool (not shown) to facilitate alignment and mounting of the guide vanes **112** to the vane housing assembly **106**.

(44) FIG. 11 is a perspective view of the first housing portion **102**. The first annular wall **126** includes a downstream surface **192** and an upstream surface **194**. The upstream surface **194** is generally annular in shape. The first annular wall **126** has a width $W_{sub.126}$ extending between the first outer surface **130** and the first inner surface **128**. (FIG. 12). The first annular wall **126** may have a height $H_{sub.126}$ of extending between the downstream surface **192** and the upstream surface **194** (FIG. 13). The first inner surface **128** defines the boundary of a first fluid flow passageway $P_{sub.122}$. The first fluid flow passageway $P_{sub.122}$ is generally cylindrical in shape having a diameter $D_{sub.122}$ defined by the first inner surface **128**. In the illustrated embodiment, the diameter $D_{sub.122}$ is substantially similar to the diameter $D_{sub.120}$ of the second inner surface **122**. (FIG. 12) The first fluid flow passageway $P_{sub.122}$ has a length $L_{sub.122}$ corresponding the height $H_{sub.122}$ of the first annular wall **126** (FIG. 13). The housing axis $A_{sub.160}$ extends through the first fluid flow passageway $P_{sub.122}$. When the first housing portion **102** and the second housing portion **104** are connected, creating the vane housing assembly **106**, the first fluid flow passageway $P_{sub.120}$ and the second fluid flow passageway $P_{sub.122}$ are aligned creating the fluid flow passageway P . The dimensions of the compressor, e.g., width $W_{sub.126}$, height $H_{sub.126}$, diameter $D_{sub.122}$, and length $L_{sub.122}$, may scaled to the size of the compressor and/or the aerodynamic needs of the compressor.

(45) FIG. 12 is a front view of the first housing portion **102**. The upstream surface **194** of the first housing portion **102** includes a plurality of first channel surfaces **196** that define a plurality of first channels **198**. The plurality of first channel surfaces **196** are arranged in a radially symmetric pattern, about the housing axis $A_{sub.106}$ that mirrors the radially symmetric pattern of the second channel surfaces **148** and the radially symmetric pattern of the plurality of guide vanes **112**. In the illustrated embodiment, each of the first channel surfaces **196** are substantially similar having approximately the same size and shape. In other example embodiments, the first channel surfaces **196** are identical to the second channel surfaces **148**. Each of the first channel surfaces **196** includes a first channel length $L_{sub.196}$ and a first channel width $W_{sub.196}$ (FIG. 12). The first channel width $W_{sub.196}$ may extend from the first inner surface **128** to the first outer surface **130**.

(46) In the illustrated embodiment, the first channel surfaces **196** are in the shape of a segment of a cylindrical surface. Accordingly, the first channels **198** are generally half-cylindrical in shape. In the illustrated embodiment, the first channel surface **196** extends along the entire width $W_{sub.126}$ of the upstream surface **194**. In other embodiments, the first channel surfaces **196** extend only partially along the width of the upstream surface **194**.

(47) As mentioned above, the stop **177** limits rotation of the guide vanes **112** about the stem axis $A_{sub.162}$ when the stop **177** engages one or more stop surfaces **195**. In the illustrated embodiment, for example, the first housing portion **102** includes a first stop surface **197** and a second stop surface **199** (FIG. 12). When the guide vane **112** rotates in a first direction (e.g., counterclockwise), the stop **177** slides within slot **154**, until the stop **177** engages the first stop surface **197**, preventing or inhibiting further rotation of the guide vane **112**. When the guide vane **112** rotates in a second direction (e.g., clockwise), the stop **177** slides within slot **154**, until the stop **177** engages the second stop surface **199**, preventing or inhibiting further rotation of the guide vane **112**.

(48) In some embodiments, contact between the stop **177** and the stop surfaces **195** serves as a stop for a stepper motor. Specifically, in the illustrated embodiment, the stop **177** is disposed within the slot **154** and when the guide vane **112** rotates, the stop **177** travels along the slot length $L_{sub.154}$. The guide vanes **112** may include a drive guide vane operably connected to a motor, and a plurality of follower guide vanes that rotate in response to rotation of the drive guide vane. In such embodiments, the stepper motor may rotate the drive guide vane **112** until the stop **177** on one of the guide vanes **112** engages with either the first stop surface **197** or the second stop surface **199**, arresting the rotational motion of all of the guide vanes **112** and stopping the stepper motor.

(49) In the illustrated embodiment, the slot **154**, the stop **177**, and the first stop surface **197** and the

second stop surface **199** are configured to allow the motor to rotate the guide vanes **112** a total of 90° . For example, the plurality of guide vanes **112** are positionable in a first rotational position of 0° , or a neutral position, such that the first vane side **166** or the second vane side **168** of each guide vane **122** is arranged generally parallel to the housing axis A.sub.106 and/or parallel to the fluid flow F entering the inlet **134**. In the illustrated embodiment, when the guide vanes **112** are in the neutral position, the stop **177** is arranged in a center of the slot **154**. The stop **177** and the stop surfaces **195** are arranged to allow the plurality of guide vanes **112** to be positionable between $+45^\circ$ and -45° degrees relative to the neutral position such that the first vane side **166** or the second vane side **168** may be arranged at a desired angle relative to the housing axis A.sub.106 and the fluid flow F entering the inlet **134** depending on the desired operating condition.

(50) In some embodiments, the second housing portion **104** can include one or more of the stop surfaces **195**. For example, in some embodiments, the first end **156** and the second end **158** includes stop surfaces that interact with stop **177**. In such embodiments, the guide vane **112** rotates, in either the first or second directions, until the stop **177** engages with the stop surfaces, preventing or inhibiting further rotation of the guide vane **112**.

(51) When the first housing portion **102** is connected to the second housing portion **104**, each of the second channels **150** are aligned with each of the first channels **198**, such that the first and second channels **198** and **150** cooperatively form a plurality of guide vane openings **200** extending radially through the vane housing assembly **106** (FIG. 2). The boundary of the guide vane openings **200** is defined by the first channel surface **196** and second channel surface **148**. Each guide vane opening **200** is generally cylindrical and is sized and shaped to receive at least a portion of the stem **162** of one of the guide vanes **112** therein. The stem **162** of each guide vane **112** is rotatable relative to the first channel surface **196** and second channel surface **148** such that each guide vane **112** rotates within one of the guide vane openings **200** about its respective stem axis A.sub.162. In some embodiments, the first channel surface **196** and the second channel surface **148** includes a plain bearing to facilitate rotation of the stem **162** relative to the first channel surface **196** and the second channel surface **148**. Additionally, and/or alternatively, the stem **162** and the first channel surface **196** and second channel surface **148** may include suitable bearings enabling the inlet guide **100** to function as described herein. In some embodiments, the stem **162** and/or the first channel surface **196** and the second channel surface **148** may be impregnated with Teflon or other suitable lubricants. In the example embodiment, the inlet guide vane **100** may be used with an oil free compressor.

(52) The first housing portion **102** and the second housing portion **104** may be connected in any suitable manner that enables the inlet guide **100** to function as described herein. In the illustrated embodiment, the first housing portion **102** is connected to the second housing portion **104** by a plurality of screws. More specifically, the first housing portion **102** includes a plurality of fastener openings **202** extending through the first annular wall **126**, and the second housing portion **104** includes a plurality of fastener opening **204** corresponding to the arrangement of the plurality of openings **202** on the first housing portion **102**. Each of the fastener openings **204** may include a threaded portion, such that a threaded bolt or screw (not shown) may be inserted through the opening **202** on the first housing portion **102** and threaded into the threaded portion of the openings **204** on the second housing portion **104**. In other embodiments, the first housing portion **102** and second housing portion **104** may be connected using any suitable fasteners to connect the first housing portion **102** and second housing portion **104** together forming the vane housing assembly **106**.

(53) When the stem **162** of each guide vane **112** is arranged within one of the guide vane openings **200** of the vane housing assembly **106**, the vane gears **164** are arranged in the exterior area **136** of the vane housing assembly **106**, and each of the plurality of vanes **160** is arranged within the fluid flow passageway P. Accordingly, in the illustrated embodiment, each of the vane gears **164** are accessible to an operator for inspection and/or repair without needing to disconnect or disassemble

the first housing portion **102** and the second housing portion **104**. In embodiments in which the vane gear **164** is removably connected to the stem **162**, the vane gear **164** can be readily replaced with another vane gear. For example, an operator may replace a worn or damaged vane gear **164** by disconnecting the vane gear **164** from the stem **162** and connecting a new or repaired vane gear **164** to the stem **162**. As mentioned above, the vane gear **164** and the stem **162** may be formed of a single piece. Furthermore, disconnecting the first housing portion **102** from the second housing portion **104** provides access to all of the plurality of guide vanes **112**, simultaneously.

(54) FIG. **14** is a perspective view of the ring gear **108**. The ring gear **108** includes a first surface **206**, a ring second surface **208**, and an annular wall **210** extending therebetween. The annular wall **210** includes a ring inner surface **212** defining the boundary of a ring opening **214**. The inner surface **212** is sized and shaped such that at least a portion of the first housing portion **102** can be received within the ring opening **214**. In the illustrated embodiment, the ring gear **108** is rotatably connected to and rotatable relative to the first housing portion **102**. In other embodiments, the ring gear **108** is rotatably connected to the second housing portion **104**.

(55) The first housing portion **102** includes a lip **216** extending radially outward from the first outer surface **130** (FIG. **13**). The lip **216** engages a surface **217** of the press fit bearing **110** when the bearing **110** is disposed around the first outer surface **130**. The lip **216** engages the bearing **110** to prevent or inhibit axial translation of the bearing **110** relative to the first housing portion **102**. Alternatively, when the ring gear **108** is rotatably connected to the first housing portion **102**, the lip **216** may be in contact with the first surface **206** of the ring gear **108** to prevent or inhibit translation axially of the ring gear **108** relative to the first housing portion **102**.

(56) FIG. **15** is an end view of the ring gear **108** and FIG. **16** is a side view of the ring gear **108**. The ring second surface **208** of the ring gear **108** includes a plurality of gear teeth **218**. The gear teeth **218** of the ring gear **108** are sized and shaped to mate with the gear teeth **186** of each of the vane gears **164** of the guide vanes **112**. Rotation of the ring gear **108**, about the housing axis A.sub.106, is transmitted to the vane gears **164**, causing rotation of the guide vanes **112** about the vane axis A.sub.162, within the guide vane openings **200** of the vane housing assembly **106**. The ring gear **108** further includes a feature **222** disposed on the inner surface **212**. The feature **222** engages with the bearing **110**, preventing the bearing **110** from translating axially relative to the ring gear **108**. The feature **222** may span about the circumference of the inner surface **212**. In some example embodiments, the feature **222** includes a lip.

(57) The ring gear **108**, and the vane gears **164**, are arranged in the exterior area **136** of the vane housing assembly **106** enabling an operator to inspect and/or repair the ring gear **108** without disconnecting the first housing portion **102** and second housing portion **104**. The accessibility of the ring gear **108** and the vane gears **164** is beneficial in that it reduces the time required to inspect and/or repair the ring gear **108** and the vane gears **164**. In addition, the accessibility of the vane gears **164** and the ring gear **108**, without disassembling the first housing portion **102** and the second housing portion **104**, helps prevent bearing surfaces between the first channel surface **196**, the second channel surface **148**, and the stem **162** from being exposed to debris and/or contaminants.

(58) In the illustrated embodiment, at least one of the guide vanes **112** is a drive guide vane **220**. The drive guide vane **220** is operably connected to motor **174** (e.g., by the driveshaft **175**), and the motor drives rotation of the drive guide vane **220**. Rotation of the drive guide vane **220** causes rotation of the ring gear **108**, which transmits rotation to the rest of the plurality of guide vanes **112**, referred to as follower guide vanes. Accordingly, all of the plurality of guide vanes **112** rotate in unison. In some embodiments, the motor is a stepper motor. In some embodiments, the motor is communicatively connected to a controller and the controller transmits one or more instructions to the motor causing the motor to rotate the drive guide vane **220** in order to arrange the plurality of guide vanes **112** in a selected orientation relative to the fluid flow F.

(59) FIG. **17** is a perspective view of the bearing **110**. The bearing **110** may be arranged between the first outer surface **130** of the first housing portion **102** and the ring inner surface **212** of the ring

gear **108**. The bearing **110** facilitates rotation of the ring gear **108** about the first housing portion **102**. In some embodiments, the bearing **110** is connected to the ring gear **108**, e.g., the bearing **110** is press fit into frictional engagement with the ring inner surface **212**. Accordingly, the ring gear **108** and the bearing **110** rotate relative to the first housing portion **102**. Alternatively, the bearing **110** may be press fit onto first housing portion **102** such that the bearing **110** and the first housing portion **102** are frictionally engaged and the ring gear **108** rotates relative to the bearing **110** and the first housing portion **102**.

(60) The bearing **110** may be a non-lubricating bearing or a self-lubricating bearing. As such, the bearing **110** does not require application of lubricants. In some embodiments, for example, the bearing **110** is constructed of bronze and/or bronze composite. In some embodiments, the bearing **110** is bronze coated. The bearing **110** may be impregnated with lubricants or the bearing **110** may include one or more graphite plugs. In alternative embodiments, the bearing **110** may include any suitable type of bearing **110** that enables the inlet guide **100** to function as described herein.

(61) In alternative embodiments, the bearing **110** may be omitted and the ring gear **108** may rotate about the first housing portion **102**, without the use of a bearing.

(62) FIG. **18** is a rear view of the inlet guide **100**, having the first housing portion **102**, ring gear **108**, and bearing **110** removed therefrom to illustrate the arrangement of the plurality of guide vanes **112** arranged within the plurality of second channels **150**. As described above, simultaneous rotation of the plurality of the guide vanes **112** changes the orientations of the vanes **160** relative to fluid flow **F** entering the inlet **134**. For example, the guide vanes **112** may be rotated, in unison, to arrange the inlet guide **100** to any suitable position based on the operational needs of the compressor. For example, the guide vanes **112** may be rotated, in unison, to arrange the inlet guide **100** to a fully open or neutral position. In the fully open position, the guide vanes **112** are arranged such that the first vane side **166** and the second vane side **168** of the vanes **160** are arranged generally parallel to the direction of the fluid flow **F** such that the vanes **160** do not substantially impede the fluid flow **F** through the fluid flow passageway **P**. In the fully open position, the vanes **160** align the fluid flow **F** creating a more laminar fluid flow **F** profile thereby increasing efficiency of the compressor. The plurality of guide vanes **112** may be rotated to arrange the vanes **160** in any suitable orientation relative to the fluid flow **F** and the fluid flow passageway **P**. For example, the motor may rotate the drive guide vane **220** in either a clockwise or a counterclockwise direction to adjust the orientation of the plurality of vanes **160**. The position of the vanes **160** may be selected to increase the operating range of the compressor, including both surge and choke.

(63) In some embodiments, sensors (not shown) may be attached to one or more of the guide vanes **112** to measure the rotational position of the guide vanes **112**. The sensor may be communicatively coupled to the controller. The controller may use feedback received from the sensor to determine instructions for the motor. In some embodiments, one or more sensors may be used to measure the rotational speed of the guide vanes **112**. For example, the guide vanes **112** may be rotated at a rotational speed of 1 rotation per min (rpm) or less.

(64) FIGS. **19** and **20** are perspective and sectional views, respectively, of a compressor **300** suitable for use with the inlet guide **100** described herein. The compressor **300** is illustrated in the form of a two-stage centrifugal compressor. The compressor **300** generally includes an outer compressor housing **302** forming at least one sealed cavity within which each stage of refrigerant compression is accomplished. The compressor **300** includes a first refrigerant inlet **310** to introduce refrigerant vapor into the first compression stage, a first refrigerant exit **314**, a refrigerant transfer conduit **312** to transfer compressed refrigerant from the first compression stage to the second compression stage, a second refrigerant inlet **318** to introduce refrigerant vapor into the second compression stage, and a second refrigerant exit (not shown in FIG. **20**). The inlet guide **100** may be arranged in proximity to the first refrigerant inlet **310**. For example, the outlet **132** to the fluid flow passageway **P** may be arranged in alignment with the first refrigerant inlet **310**. Refrigerant transfer conduit **312** is operatively connected at opposite ends to the first refrigerant exit **314** and

the second refrigerant inlet **318**, respectively. The second refrigerant exit delivers compressed refrigerant from the second compression stage to a cooling system in which compressor **300** is incorporated. The refrigerant transfer conduit **312** may further include a refrigerant bleed (not shown in FIG. **20**) to add or remove refrigerant as needed at the compressor **300**.

(65) Referring to FIG. **20**, the outer compressor housing **302** encloses a first compression stage **324** and a second compression stage **326** at opposite ends of the compressor **300**. The first compression stage **324** includes a first stage impeller **306** configured to impart kinetic energy to incoming refrigerant gas entering via the first refrigerant inlet **310**. The kinetic energy imparted to the refrigerant by the first stage impeller **306** is converted to increased refrigerant pressure (i.e., compression) as the refrigerant velocity is slowed upon transfer to a diffuser formed between a first stage inlet ring **301** and a portion of the outer compressor housing **302**. Similarly, the second compression stage **326** includes a second stage impeller **316** configured to add kinetic energy to refrigerant transferred from the first compression stage **324** entering via the second refrigerant inlet **318**. The kinetic energy imparted to the refrigerant by the second stage impeller **316** is converted to increased refrigerant pressure (i.e., compression) as the refrigerant velocity is slowed upon transfer to a diffuser formed between a second stage inlet ring **303** and a second portion of outer compressor housing **302**. Compressed refrigerant exits the second compression stage **326** via the second refrigerant exit (not shown in FIG. **20**).

(66) The first stage impeller **306** and second stage impeller **316** are connected at opposite ends of a driveshaft **304** that rotates about a driveshaft axis A.sub.304. The driveshaft extends from a driveshaft first end **330** to a driveshaft second end **332** and is axisymmetric about the driveshaft axis A.sub.304. Additionally, the driveshaft axis A.sub.304 extends through a center of gravity of the driveshaft **304**. The driveshaft **304** is operatively connected to a motor **308** positioned between the first stage impeller **306** and second stage impeller **316**, such that the motor **308** rotates the driveshaft **304** about the driveshaft axis A.sub.304. The first stage impeller **306** and the second stage impeller **316** are both connected to the driveshaft **304** such that the first stage impeller **306** and second stage impeller **316** are rotated at a rotation speed selected to compress the refrigerant to a pre-selected pressure exiting the second refrigerant exit. Any suitable motor may be incorporated into the compressor **300** including, but not limited to, an electrical motor.

(67) Embodiments of the inlet guide vane apparatus described herein require simple assembly of a few number of components. Embodiments of the inlet guide vane apparatus, described above, include a first and second housing portion, a ring gear and bearing, and a plurality of guide vanes. In the illustrated embodiments, the vane gear of the inlet guide vane and the ring gear are both arranged in an area exterior to the vane housing. Accordingly, these components are accessible to an operator to perform and inspection and/or a repair operation. Specifically, the operator does not need to disassemble the inlet guide vane apparatus to inspect and/or replace the gears of the guide vanes. In one illustrated embodiment, the vane gear is selectively connected to the stem of the guide vane. Accordingly, the vane gear may be removed and replaced, without requiring the operator to disassemble the housing minimizing exposure of bearing surfaces to contaminants. Embodiments of the bearing may be a non-lubricating bearing or is a self-lubricating bearing. As such, the bearing does not require application of lubricants for operation. Embodiments described herein include a relatively small number of components, e.g., a two part “split-housing” that allows the operator to quickly (e.g., within 10 minutes) disassemble and/or reassemble the inlet guide vane apparatus. Furthermore, an operator may only need to use a single tool to disassemble and/or reassemble the inlet guide vane.

(68) As used herein, the terms “about,” “substantially,” “essentially” and “approximately” when used in conjunction with ranges of dimensions, concentrations, temperatures or other physical or chemical properties or characteristics is meant to cover variations that may exist in the upper and/or lower limits of the ranges of the properties or characteristics, including, for example, variations resulting from rounding, measurement methodology or other statistical variation.

(69) When introducing elements of the present disclosure or the embodiment(s) thereof, the articles “a”, “an”, “the” and “said” are intended to mean that there are one or more of the elements. The terms “comprising,” “including,” “containing” and “having” are intended to be inclusive and mean that there may be additional elements other than the listed elements. The use of terms indicating a particular orientation (e.g., “top”, “bottom”, “side”, etc.) is for convenience of description and does not require any particular orientation of the item described.

(70) Having thus described several illustrative embodiments, it is to be appreciated that various alterations, modifications, and improvements will readily occur to those skilled in the art. Such alterations, modifications, and improvements are intended to form a part of this disclosure, and are intended to be within the spirit and scope of this disclosure. While some examples presented herein involve specific combinations of functions or structural elements, it should be understood that those functions and elements may be combined in other ways according to the present disclosure to accomplish the same or different objectives. In particular, acts, elements, and features discussed in connection with one embodiment are not intended to be excluded from similar or other roles in other embodiments. Additionally, elements and components described herein may be further divided into additional components or joined together to form fewer components for performing the same functions. Accordingly, the foregoing description and attached drawings are by way of example only and are not intended to be limiting.

Claims

1. An inlet guide vane apparatus, the apparatus comprising: a housing assembly defining a fluid flow passageway, the housing assembly comprising: a first housing portion; and a second housing portion connected to and positioned axially upstream of the first housing portion, wherein the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway, wherein at least one of the first housing portion and the second housing portion defines at least one slot, the slot extending from a first stop surface to a second stop surface; a ring gear rotatable relative to the housing assembly; and a plurality of guide vanes connected to the housing assembly, each guide vane comprising: a stem extending from a first end to a second end, wherein at least a portion of the stem is disposed within one of the guide vane openings, wherein at least one of the guide vanes includes a stop extending from the stem and being sized and shaped to be received in the slot; a vane gear disposed at the first end of the stem and operably connected to the ring gear; and a vane disposed at the second end of the stem and within the fluid flow passageway of the housing assembly, wherein each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.
2. The inlet guide vane apparatus of claim 1, wherein the ring gear and the vane gear are accessible from an exterior of the housing assembly.
3. The inlet guide vane apparatus of claim 1, wherein the first housing portion defines a plurality of first channels and the second housing portion defines a plurality of second channels, wherein each of the first channels is aligned with one of the second channels such that the first and second channels cooperatively define the plurality of guide vane openings.
4. The inlet guide vane apparatus of claim 1 further comprising a bearing, wherein the ring gear is rotatably connected to the housing assembly by the bearing.
5. The inlet guide vane apparatus of claim 4, wherein the bearing is a self-lubricating bearing.
6. The inlet guide vane apparatus of claim 1, wherein the plurality of guide vanes includes a drive guide vane and a plurality of follower guide vanes.
7. The inlet guide vane apparatus of claim 6, wherein the drive guide vane is operably connected to a motor operable to rotate the drive guide vane, wherein rotation of the drive guide vane drives rotation of the ring gear and rotation of the ring gear imparts rotation to the plurality of follower

guide vanes.

8. The inlet guide vane apparatus of claim 7, wherein the motor is a stepper motor, and wherein contact between the stop and at least one of the first and second stop surfaces stops rotation of the plurality of guide vanes and stops the motor.

9. The inlet guide vane apparatus of claim 1, wherein each guide vane includes the stop extending radially outward from the stem.

10. The inlet guide vane apparatus of claim 1, wherein the vane gear of each guide vane is removably connected to the guide vane.

11. A compressor comprising: a compressor housing including an inlet; a driveshaft rotatably supported within the compressor housing; an impeller connected to the driveshaft and operable to impart kinetic energy to incoming refrigerant gas upon rotation of the driveshaft; an inlet guide vane apparatus connected to the compressor inlet, the inlet guide vane apparatus comprising: a housing assembly defining a fluid flow passageway, the housing assembly comprising: a first housing portion; and a second housing portion connected to and positioned axially upstream of the first housing portion, wherein the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and into the fluid flow passageway, wherein at least one of the first housing portion and the second housing portion defines at least one slot, the slot extending from a first stop surface to a second stop surface; a ring gear rotatable relative to the housing assembly; and a plurality of guide vanes connected to the housing assembly, each guide vane comprising: a stem extending from a first end to a second end, wherein at least a portion of the stem is disposed within one of the guide vane openings, wherein at least one of the guide vanes includes a stop extending from the stem and being sized and shaped to be received in the slot; a vane gear disposed at the first end of the stem and operably connected to the ring gear; and a vane disposed at the second end of the stem and within the fluid flow passageway of the housing assembly, wherein each of the guide vanes is rotatable relative to the housing assembly such that an orientation of the vane within the fluid flow passageway is selectively adjustable.

12. The compressor of claim 11, wherein the ring gear and the vane gear are accessible from an exterior of the housing assembly.

13. The compressor of claim 11, wherein the first housing portion defines a plurality of first channels and the second housing portion defines a plurality of second channels, wherein each of the first channels is aligned with one of the second channels such that the first and second channels cooperatively form the plurality of guide vane openings.

14. The compressor of claim 11, further comprising a bearing, wherein the ring gear is rotatably connected to the housing assembly by the bearing.

15. The compressor of claim 14, wherein the bearing is a self-lubricating bearing.

16. The compressor of claim 11, wherein the plurality of guide vanes includes a drive guide vane and a plurality of follower guide vanes.

17. The compressor of claim 16, wherein the drive guide vane is operably connected to a motor operable to rotate the drive guide vane, wherein rotation of the drive guide vane drives rotation of the ring gear and rotation of the ring gear imparts rotation to the plurality of follower guide vanes.

18. The compressor of claim 17, wherein the motor is a stepper motor, and wherein contact between the stop and at least one of the first and second stop surfaces stops rotation of the plurality of guide vanes and stops the motor.

19. A method of assembling an inlet guide vane apparatus including a housing assembly and a plurality of guide vanes, wherein the housing assembly includes a first housing portion and a second housing portion, and wherein each guide vane includes a stem extending from a first end to a second end, a first vane gear disposed at the first end of the stem, and a vane disposed at the second end of the stem, and wherein at least one of the guide vanes includes a stop extending from the stem, the method comprising: rotatably connecting a ring gear to the first housing portion;

connecting the first housing portion to the second housing portion such that the second housing portion is positioned axially upstream of the first housing portion, and such that the first and second housing portions cooperatively define a plurality of guide vane openings extending radially through the housing assembly and wherein at least one of the first housing portion and the second housing portion defines at least one slot; and positioning the plurality of guide vanes relative to the second housing portion such that, when the first housing portion is connected to the second housing portion, at least a portion of the stem of each guide vane is disposed within one of the guide vane openings and the stop is disposed within the slot.

20. The method of claim 19, wherein the first vane gear of each guide vane is removably connected to the stem, wherein the method further comprises: removing the first vane gear from the stem of one of the guide vanes; and connecting a second vane gear to the stem of the one guide vane, without disconnecting the first housing portion from the second housing portion.
